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Physics-Informed Neural Network for a Reduced WNT-HOX-Retinoic Acid Model of Colorectal Cancer

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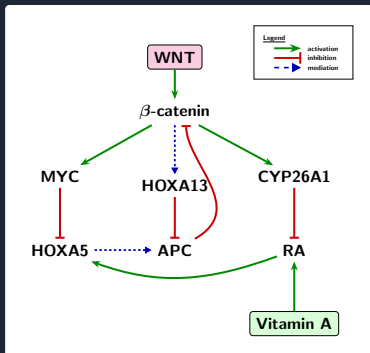
Overview

- Biology/Schematic
- Model
- Forward PINN
- Inverse PINN
- Inverse Bayesian PINN
- Next Steps

Biological Network Governing WNT–HOX–RA Signaling

Objective

Develop a reduced mathematical model describing the interactions among the **WNT**, **HOX**, and **Retinoic Acid (RA)** pathways during colorectal cancer progression.



State Variables

$y =$

$[B(t), P(t), H_5(t), H_{13}(t), M(t), R(t), C(t)]$

B: β -catenin

P: APC

H₅: HOXA5

H₁₃: HOXA13

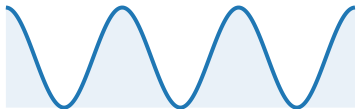
M: MYC

R: Retinoic Acid

C: CYP26A1

RA Input + Treatment

RA Input



ATRA Treatment



$$A_d \left[1 + \cos \left(\frac{2\pi t}{T_d} - \phi \right) \right]$$

$$u_R(t) = u_0 + \frac{D}{2} [\tanh(q(t - t_1)) - \tanh(q(t - t_2))].$$

Dimensional Model

$$\frac{dB}{dt} = a_B W + v_{13} \frac{H_{13}^{n_H}}{K_{13}^{n_H} + H_{13}^{n_H}} - d_B B - k_{PB} PB - v_5 \frac{H_5 B}{K_5 + B}$$

$$\frac{dP}{dt} = a_P \frac{1 + \rho_5 H_5}{1 + \rho_B B + \rho_{13} H_{13}} - (d_{P0} + d_{P1}(1 - \Theta_P)) P$$

$$\frac{dH_5}{dt} = a_5 + v_R \frac{R}{K_R + R} - d_5 H_5 - v_M \frac{M H_5}{K_M + M}$$

$$\frac{dH_{13}}{dt} = a_{13} + v_{B13} \frac{B^{n_B}}{K_{B13}^{n_B} + B^{n_B}} + v_{M13} \frac{M^{n_M}}{K_{M13}^{n_M} + M^{n_M}} - d_{13} H_{13}$$

$$\frac{dM}{dt} = a_M + v_{BM} \frac{B^{n_B}}{K_{BM}^{n_B} + B^{n_B}} - d_M M$$

$$\frac{dR}{dt} = u_0 + A_d \left[1 + \cos \left(\frac{2\pi t}{T_d} - \phi \right) \right] + \frac{D}{2} \left[\tanh(q(t - t_1)) - \tanh(q(t - t_2)) \right] - d_R R - k_{CR} CR$$

$$\frac{dC}{dt} = a_C + v_{RC} \frac{R}{K_{RC} + R} + v_{BC} \frac{B^{n_B}}{K_{BC}^{n_B} + B^{n_B}} - d_C C$$

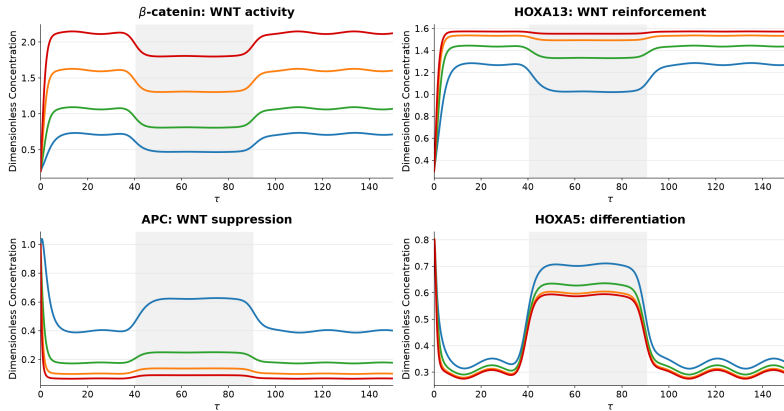
4 Regimes

Disease progression is modeled by increasing WNT signaling (W) and progressive loss of APC function (Θ_p), mimicking CRC progression.

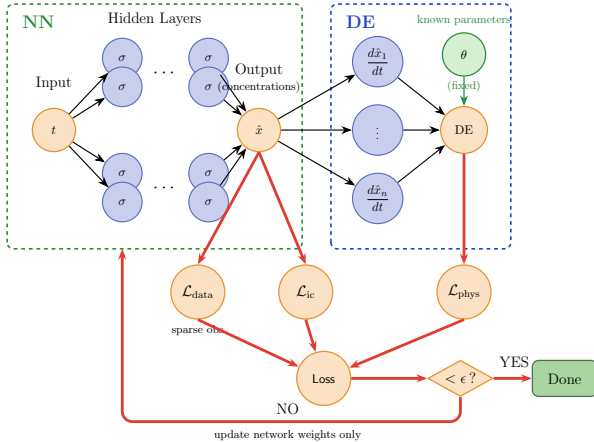
- “Normal”
 - $W = 0.8$, $\Theta_p = 1.0$
- “Early Adenoma”
 - $W = 1.0$, $\Theta_p = 0.75$
- “Advanced Adenoma”
 - $W = 1.5$, $\Theta_p = 0.5$
- “Severe APC Loss”
 - $W = 2.0$, $\Theta_p = 0.25$

Core WNT-HOX Regulatory Dynamics

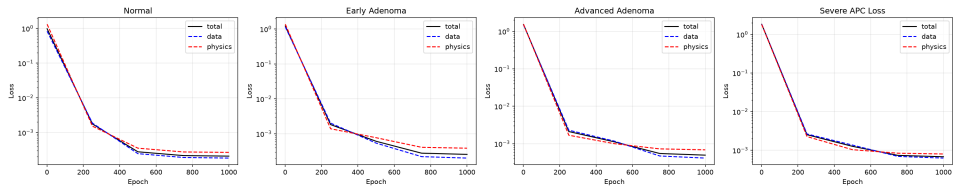
Normal Early Adenoma Advanced Adenoma Severe APC Loss ATRA Window



Forward PINN

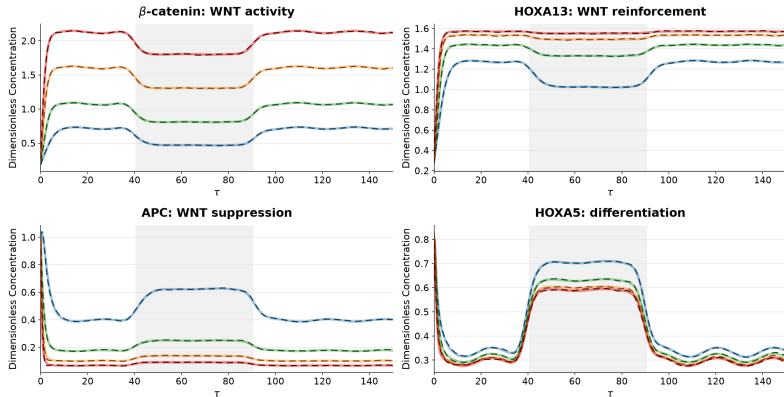


Forward PINN Loss

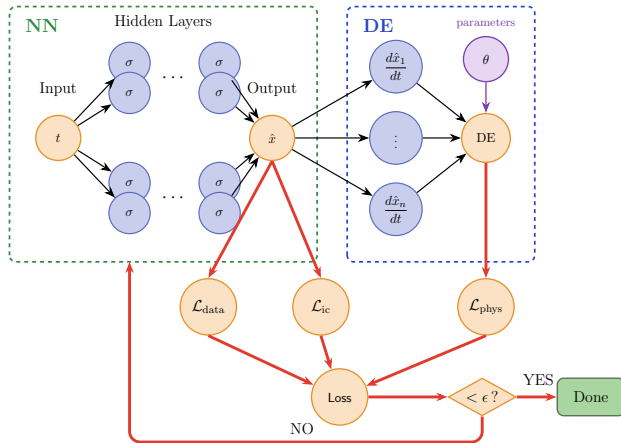


Core WNT-HOX Regulatory Dynamics via Forward PINN

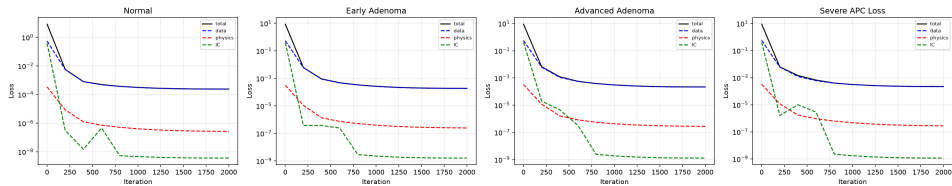
— Normal (PINN) — Early Adenoma (PINN) — Advanced Adenoma (PINN) — Severe APC Loss (PINN) ATRA Window
— Normal (ODE solver) — Early Adenoma (ODE solver) — Advanced Adenoma (ODE solver) — Severe APC Loss (ODE solver)



Inverse PINN

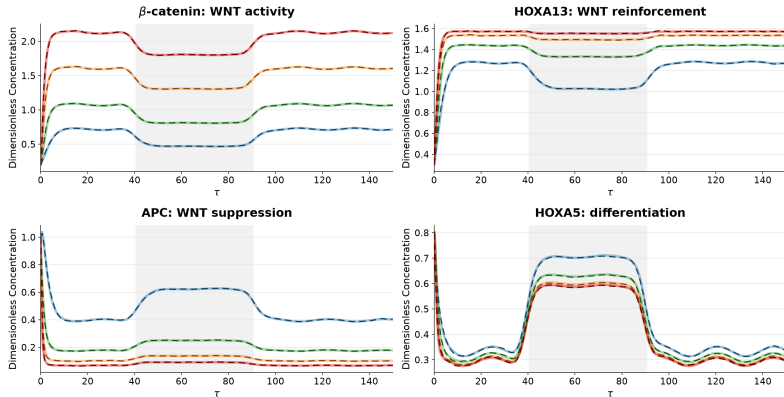


Inverse PINN Loss

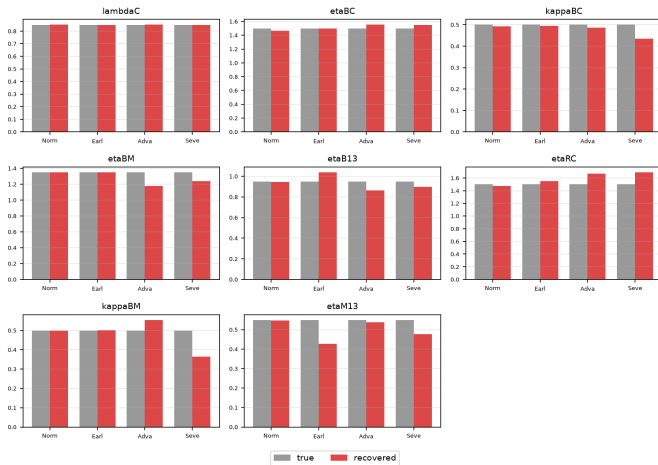


Core WNT-HOX Regulatory Dynamics via Inverse PINN

— Normal (PINN) — Early Adenoma (PINN) — Advanced Adenoma (PINN) — Severe APC Loss (PINN) ATRA Window
— Normal (ODE solver) — Early Adenoma (ODE solver) — Advanced Adenoma (ODE solver) — Severe APC Loss (ODE solver)

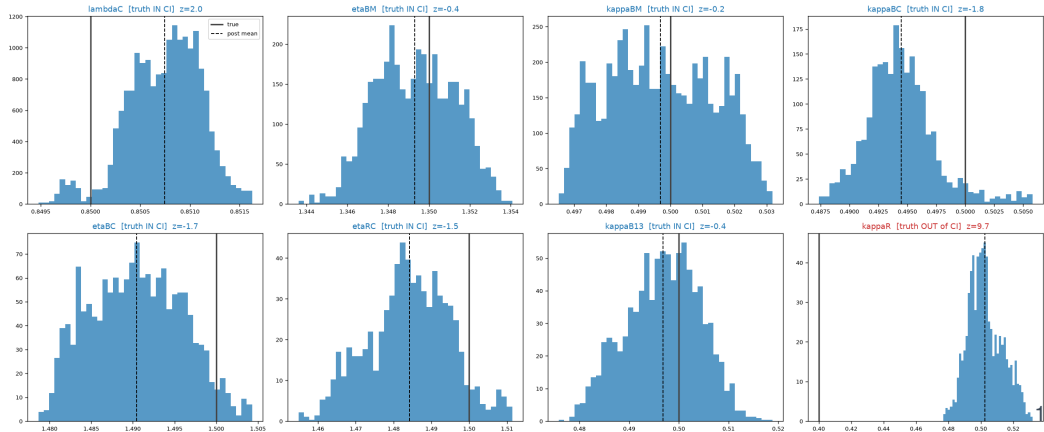


Inverse PINN — true vs recovered (8 best-converging params)



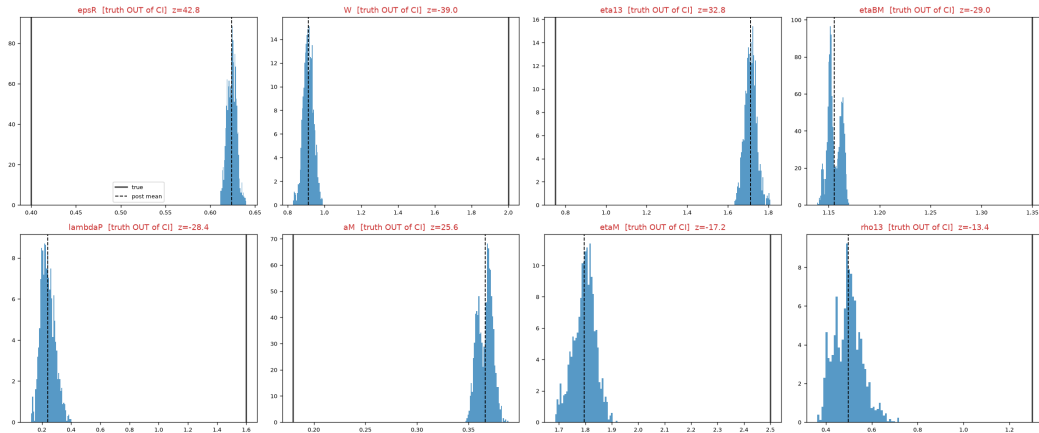
Inverse Bayesian PINN — uncertainty quantification

Normal — posterior marginals (top-8 most sensitive) [blue = truth inside 95% CI, red = confidently wrong] coverage 18/36



Inverse Bayesian PINN — continued

Severe APC Loss — posterior marginals (8 worst-recovered) [blue = truth inside 95% CI, red = confidently wrong] coverage 13/36



Parameter Recovery Comparison

		Bayesian inverse PINN	Inverse PINN	ODE-fit
“Normal”		19	17	18/36
“Early Adenoma”		20	16	17/36
“Advanced	Ade-	11	10	13/36

Next Steps

- Improve Bayesian PINN
 - Investigate limitations
- Hybrid Mechanistic-Neural Model
- Stochastic Modeling

Next Steps graphic
missing

Key References

1. **Raissi, Perdikaris & Karniadakis (2019).** Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear PDEs. *J. Comput. Phys.* **378**, 686–707.

Foundational PINN formulation — the data + physics-residual loss used in every model here.

2. **Yazdani, Lu, Raissi & Karniadakis (2020).** Systems biology informed deep learning for inferring parameters and hidden dynamics. *PLoS Comput. Biol.* **16**(11), e1007575.

Closest template to this work: inverse PINNs on biological ODE networks; the local-sensitivity/Fisher route to identifiability.

3. **Yang, Meng & Karniadakis (2021).** B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data. *J. Comput. Phys.* **425**, 109913.

Basis of the Bayesian inverse PINN: HMC over parameters gives posterior marginals instead of a point estimate.

4. **Raue et al. (2009).** Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics* **25**(15), 1923–1929.

Thank you!

Questions? (we will try to answer)

PK on the phone
missing

PK calling Amir + Nate to check in on our progress...

Nate + diet coke
missing

4th diet coke of the day +
Celsius b/c his sleep
schedule is so cooked

Amir, world cup
missing

Don't know what in the
world is going on b/c he
spent more hours watching

group photo
missing

Thank you very much to the
Provost's Office for funding our
summer research project!